

$$= 2^{\log n} + n \log n$$

$$= n + \frac{n \log n}{\downarrow \text{is big}}$$

$$O(n \log n)$$

Masters theorem for dividing functions

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$$a \geq 1 \quad b \geq 2 \quad f(n) = \Theta(n^k \log^p n)$$

Case 1 if $\log_b a > k$ then $\Theta(n^{\log_b a})$

Case 2 if $\log_b a = k$

if $p > -1$ then $\Theta(n^k \log^{p+1} n)$

if $p = -1$ then $\Theta(n^k \log \log n)$

if $p < -1$ then $\Theta(n^k)$

Case 3 if $\log_b a < k$

if $p \geq 0$ then $\Theta(n^k \log^p n)$

if $p < 0$ then $O(n^k)$

ex 1 of case 1

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$$a=2 \quad b=2$$

$$f(n) = \Theta(1) = \Theta(n^0 \log^0 n)$$

① $\log_b a = \log_2 2 = 1$

② $k = 0$

① > ②

$$\Theta(n^1)$$

ex 2

$$T(n) = 4T\left(\frac{n}{2}\right) + n$$

① $\log_2 4 = 2$

② $k = 1$

① > ②

$$\Theta(n^2)$$

ex 3

$$8T\left(\frac{n}{2}\right) + n$$

① $\log_2 8 = 3$

② $k = 1$

① > ②

$$\Theta(n^3)$$

ex 4

$$aT\left(\frac{n}{3}\right) + 1$$

① $\log_3 a = 2$

② $k \geq 0$

① > ②

$$\Theta(n^2)$$

ex 1 of case 2

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

① $\log_2 2 = 1$

② $k = 1$

① = ②

$p = 0$ as no log

$$\Theta(n^1 \log^{0+1} n)$$

$$\Theta(n \log n)$$

if ① = ②

then multiply last with log n

③ $4T\left(\frac{n}{2}\right) + n^2$

① $\log_2 4 = \log_2 2^2 = 2$

② $k = 2$

① = ②

$$\Theta(n^2 \log n)$$

③ $4T\left(\frac{n}{2}\right) + n^2 \log n$

$$O(n^2 \log n \log n)$$

④ $4T\left(\frac{n}{2}\right) + n^2 \log^2 n$

$$O(n^2 \log^3 n)$$

⑤ $8T\left(\frac{n}{2}\right) + n^3$

$$\log_2 8 = 3$$

$$k = 3$$

$$O(n^3 \log n)$$

ex 2 of case 2

$$T(n) = 2T\left(\frac{n}{2}\right) + \frac{n}{\log n}$$

① $\log_2 2 = \log_2 2 = 1$

② $k = 1$

① < ②

$$\Theta(n \log n \log n)$$

(c) $\frac{n}{\log^2 n}$

$$= n^1 \log^{-2} n$$

$p = -2$

$$\Theta(n)$$

ex 3

$$T(n) = T\left(\frac{n}{2}\right) + n^2$$

① $\log_2 1 = 0 = 2 \times 0 = 0$

② $k = 2$

① < ②

$$\Theta(n^2)$$

③ $T(n) = 2T\left(\frac{n}{2}\right) + n^2 \log n$

① $\log_2 2 = 1$

② $k = 2$

① < ②

$$\Theta(n^2 \log n)$$

$$O(n)$$

$$O(n \log n)$$

(b) $T(n) = 4T(n/2) + \frac{n^3}{\log n}$

① $\log_2 2 = 2$

② $k = 3$

③ $p = 3 > 0$

$O(n^3)$

any log in denominator then don't take it.